A Paper Currency Recognition System Using Negatively Correlated Neural Network Ensemble

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Abstract—**This paper presents a currency recognition system using ensemble neural network (ENN). The individual neural networks (NNs) in an ENN are trained via negative correlation learning. The objective of using negative correlation learning (NCL) is to expertise the individuals on different parts or portion of input patterns in an ensemble. The available currencies in the market consist of new, old and noisy ones. It is often difficult for machine to recognize these currencies; therefore we propose a system that uses ENN to identify them efficiently. We performed our experiment for seven different types of TAKA (Bangladeshi currency) which are 2, 5, 10, 20, 50, 100 and 500 TAKA. The image of different types note is converted in gray scale and compressed in the desired range. Each pixel of the compressed image is given as an input to the network. This system is able to recognize highly noisy or old image of TAKA. Ensemble network is very useful for the classification of different types of currencies. It reduces the chances of misclassification than a single network and ensemble network with independent training. In experimental results we have shown this with artificially added noise in the test set. We also find good result for similar pattern available in market.**

Index Terms— **Neural network, Neural network ensemble, Diversity, Negative correlation learning, Image processing, Currency recognition**

I. Introduction

Artificial neural networks (ANNs) have been applied in various application domains for solving real world problems such as, feature extraction from complex data sets, direct and parallel implementation of matching and search algorithm, forecasting and prediction in a rapidly changing environment, recognition and image processing applications etc. The currency recognition is one of the important application domains of ANN.

A reliable currency recognition system is important for the automation in different sectors for a country such as vending machine, rail way ticket counter, banking system, shopping mall, currency exchange service etc. The paper currency recognition is important for a number of reasons. a) They become old frequently than coins; b) The possibility of joining broken currency is higher than that of coin currency; c) Coin currency is limited to smaller range.

Many kinds of bank note recognition machines are available in our society. They are very useful devices to make people free from difficulties in counting bank notes, changing money or vending tickets. On the other hand, it is true that the recent advancement of copy machines or scanners enables us to duplicate counterfeit bank notes, although there is machine and/or scanners to reject those banknotes [1]. However, old machines do not have this facility. Therefore, a reliable currency recognition system is indeed necessary.

Various methodologies have been proposed for the recognition of paper currency in different countries. Most of these use a single feed-forward neural network (NN) for the recognition [2-6]. Takeda used a neural network model to paper currency recognition [7]. The features are first extracted from the image, and then they are applied to the network for training. It uses the edge detection technique for feature extraction [2] and it reduces the network size. For the new notes, feature extraction from the edge detection is easy, but for the noisy notes, it is very difficult. Furthermore the currency recognition system should be highly reliable. If a network makes a false classification, it will be impractical, so single neural network is not reliable enough. Therefore we propose 'ensemble neural network' (ENN) [8] to address this problem. The ensemble NN can solve the problem efficiently than single NN. The 'negative correlation learning' (NCL) is used to produce different individual NN in the ensemble, so that the entire ensemble learns the input patterns completely.

In our proposed system we use the ENN for currency recognition. NCL was used for the training of the ensemble network [9]. The use of NCL is to produce the diversity among the individual networks in the ensemble. The final decision of the network is taken from voting among the individual NN. In voting each network gives a vote for a certain class and it is done by the winning neuron of that network. Our experimental result is very much good for the recognition of noisy images and found

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